Table 5

Binary attack-detection performance and decision-threshold characteristics

A. Threshold-independent discrimination and probability calibration

Attack AP	ROC-AUC	Brier score	ECE (10 bins)	Attack prevalence
0.9967	**0.9929**	**0.0309**	**0.0055**	**0.6818**

B. Operating characteristics at fixed attack thresholds

Threshold	Precision (%)	Recall (%)	Binary F1 (%)	FPR (%)	False positives	Missed attacks
0.50	96.00	97.41	96.70	8.69	973	621
0.70	98.12	93.73	95.87	3.84	430	1,504
0.80	99.03	90.48	94.56	1.90	213	2,283
0.90	99.64	86.57	92.65	0.66	74	3,219
0.95	99.84	83.20	90.77	0.28	31	4,027

Notes: AP = average precision; ROC-AUC = area under the ROC curve; ECE = expected calibration error; FPR = false-positive rate.
Threshold-specific measurements classify Attack when 1 - P(Normal) is at least the stated threshold. Counts refer to network-flow records.
UNSW-NB15 development validation: 35,167 records (23,976 Attack; 11,191 Normal). Scores were reproduced from the saved XGBoost V1 classifier.
Development validation was repeatedly examined; the results do not estimate independent TEST-set performance or real-world deployment reliability.